

PROJECT REPORT

ON

Grace Care

By

KRISH PADSALA (22012250410144)

SHAIKH MOHAMMED HARIS (22012250410186)

MOMIN MOHAMMAD ALI (22012250410190)

SAIYED MOHAMMED ANISH (22012250410192)

SHAIKH MOHAMMED TAMIM (22012250410194)

QURESHI SAFWAN (22012250410267)



DEPARTMENT OF INFORMATION TECHNOLOGY

L.J.POLYTECHNIC, AHMEDABAD

2024-2025

DEPARTMENT OF INFORMATION TECHNOLOGY

L.J.POLYTECHNIC, AHMEDABAD

2024-2025

CERTIFICATE

Date: ____/____/____

This is to certify that **Mr. KRISH PADSALA, Mr. SHAIKH MOHAMMED HARIS, Mr. MOMIN MOHAMMAD ALI, Mr. SAIYED MOHAMMED ANISH, Mr. SHAIKH MOHAMMED TAMIM And Mr. QURESHI SAFWAN** from LJ POLYTECHNIC having Enrollment No. **22012250410144, 22012250410186, 22012250410190, 22012250410192, 22012250410194, 22012250410267** have completed project documentation and development on the problem definition of semester VI during the academic year 2024-25 having Title **Grace Care** in a group consisting of 6.

Institute Guide

Head of the Department

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KRISH PADSALA	(22012250410144)
SHAIKH MOHAMMED HARIS	(22012250410186)
MOMIN MOHAMMAD ALI	(22012250410190)
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SHAIKH MOHAMMED TAMIM	(22012250410194)
QURESHI SAFWAN	(22012250410267)

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ABSTRACT

We are developing a system grace Care regarding hospital Management. Our Grace Care-Hospital Management System is designed primarily for our hospitals healthcare professionals including, doctor and patient. With This System we want to make things simpler and more efficient. This system will help to keep Record of Doctors and Patient Easily. It will streamline operations by allowing admins to manage staff details, enabling doctors to view their appointments and patient histories, and making it easy for patients to book appointments. This upgrade will enhance accuracy, speed and communication, leading to a more easy and efficient organized workflow. We can keep track of appointments better and make sure that everyone has the information they need it.

CHAPTER-1

INTRODUCTION

1.1 Need of the system

- Our current manual system at the hospital is slow and prone to mistakes, making it hard to keep up with patient records and appointments.
- With this system we want to make things simpler and more efficient.
- By having everything in one place, we can avoid errors, keep track of appointments better, and make sure everyone has the information they need when they need it.

1.2 Detailed problem definition

- The current Hospital Management System causes problems like find the patient record, his medical history, taking a long time to find information, and making more mistakes.
- This Hospital Management system is here to fix those problems by giving us a trustworthy, fast, and safe digital place to work.
- The Learn Together platform solves these issues with three main components:
 1. **Admin Panel:** Admins can manage Doctor, patient and Staff Records.
 2. **Patient Panel:** Patient can see Doctor List, Book Appointment. His previous and current appointment.

3. **Doctor Panel:** Doctor can Register Patient, He Able to see Patient List and Able to manage medical history of patient which is his under. Doctor Can see all Appointment which is h0FRis under and able to approve or cancel.

1.3 Viability of the system

- **Technical Feasibility:** Can the system be built and maintained using available technology. It should run smoothly on the necessary hardware and software platforms.
- **User-Friendliness:** Is the system easy for doctors, nurses, admin staff, and patients to use? A system that is too complex could discourage use.
- **Scalability:** Can the system handle a growing number of patients, staff, and data as the hospital expands.
- **Interoperability:** Can the system integrate with other existing software, such as lab information systems or pharmacy management tools.

1.4 Presently available system

- <https://www.medicalsoftwaresystem.com>
- <https://www.practo.com/hospitals>
- Many records are still maintained manually.
- Existing systems are not highly user-friendly.
- Current systems may be limited to standalone or local network implementations.

1.5 Future prospects

- To enhance our Hospital Management System, we can introduce a mobile app for patients to access their medical records, check appointment schedules, and receive reminders.
- This will improve patient engagement and convenience. Additionally, integrating AI-powered chatbots for answering common queries and automating administrative tasks like appointment reminders can further streamline operations and improve overall efficiency.

2.1. Requirement Analysis

2.1.1 Functional Requirements:

- **Registration:**
 - The admin is the only user who can register doctors. The patient can register through the registration module.
- **Appointment Scheduling:**
 - Functionality to schedule, reschedule, and cancel appointments for patients with healthcare professional.
- **Patient Panel:**
 - Patient login and profile management.
 - View Appointment records.
- **Admin Panel:**
 - Manage Nurse and Doctors Generate reports and analytics.
 - Add Doctors and Nurse information.
- **Doctor Panel:**
 - Doctor login and profile management.
 - Mark Appointment and upload patient records.
 - Access patient information.
- **Authentication**
 - **Login** – Users (Admin, Doctor, and patient) will log into the system using their unique username and password.

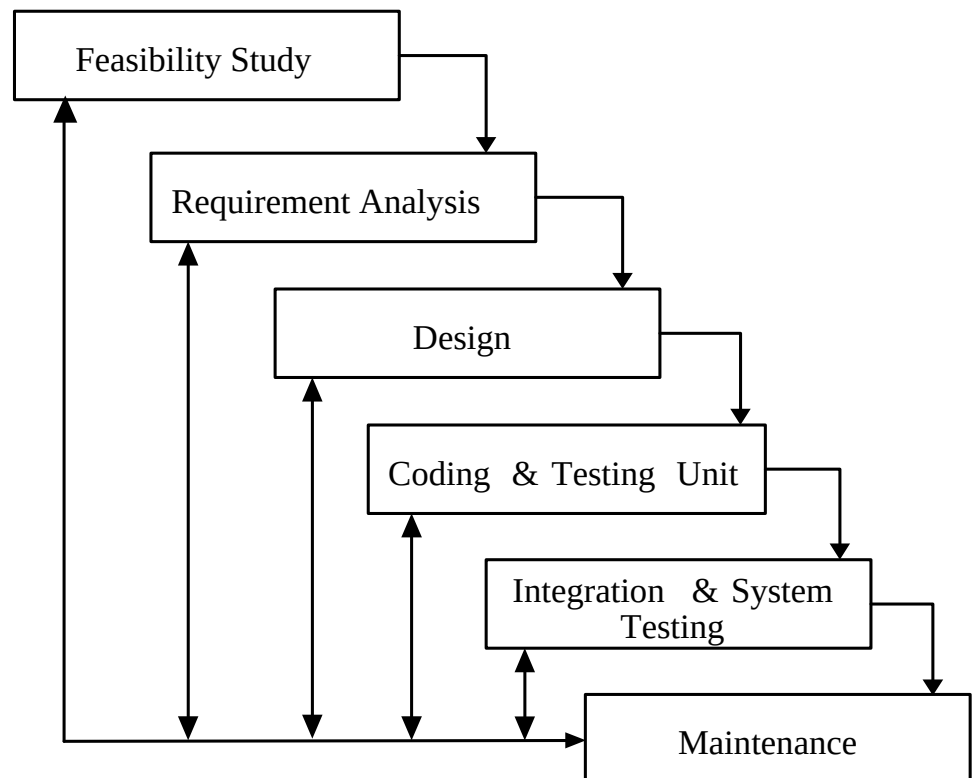
- **Logout** – After completing their requirement, users can log out securely.
- **Forget Password** – If a user (Doctor and patient) has forgotten his password he can tell the admin that I have forgotten my password so that the admin can provide him with a new password.
- **Process data**
 - **Display** – Users with defined roles can access relevant data based on their permissions.
 - **Doctor:** Doctor can register patient (add), manage appointment (approve, cancel) and discharge patient.
 - **Patient:** Can view their data such as appointment, medical history and also patient cancel his appointment.
 - **Edit** – Editing rights are assigned based on user roles to ensure data integrity.
 - **Admin:** it can add, update and remove the detail of doctor and patient.
 - **Data Security and Privacy** – Data displayed and processed will be protected through role-based access control, ensuring that each user only accesses the information they are authorized to view.

2.1.2 Non-Functional Requirements:

- **Security:**
 - User authentication and authorization.
 - Data encryption and backup.
- **Usability:**
 - User-friendly interface and navigation.
 - Clear instructions and help resources.
- **Performance:**

- Fast loading times and responsive design.
- **Compatibility:**
 - Cross-browser compatibility
 - Mobile device accessible

2.2 Project Model



[Figure 1: Iterative Waterfall Model]

- This application is developed using Iterative model. Almost every other model is derived from the waterfall model.
- The phase of detecting errors is close to its points of introduction is known as face containment of errors.
- Incremental model is also referred as the successive version of waterfall model using incremental approach and evolutionary model.
- In this model, the system has broken down into several modules which can be incrementally implemented and delivered.
- First develop the core model and when customer evaluate the system then the initial product skeleton is redefined into increasing levels capacity by adding new functionalities in successive versions.
- **Advantages**
 - Each successive version performing more useful work than previous versions.
 - The core modules get tested thoroughly, thereby reducing change of error in final product.
 - The model is more flexible and less costly to change the scope and requirement.
 - User gets a change to experiment with partially developed software.
 - This model helps finishing exact user requirements.
 - Feedback providing at each increment is useful for determining the better final product.

2.3 Schedule Representation

- Generalized project scheduling tools and technique can be applied with little modification to software projects.
- Project evolution and review technique and critical paths method are two project scheduling method that can be applied to software development.
- Both techniques are driven by information already developed in earlier project planning activities.
- A decomposition of the product function.
- The selection of appropriate process model and task set. Estimate of effort.
- Decomposition of data.

[Table 1: Schedule Representation]

ACTIVITY	START DATE	FINISH DATE
Requirement Analysis		
System Analysis		
System Design		
System Coding		
Testing and Integration		

2.4. Feasibility Study:

2.4.1 Technical Feasibility:

- The Hospital Management System is a web-based application designed to manage and streamline various Hospital operations, including Appointment

management, Patient management, Patient panel, admin panel, and Doctor Panel.

- Programming Languages: Core php.
- Database: Core php, MySQL.
- Front-end: HTML, CSS, and JavaScript.

2.4.2 Economical Feasibility:

- It looks at whether the hospital can afford the initial investment, maintenance costs, and staff training, while ensuring the system improves efficiency, reduces operational costs, and ultimately provides value.
- **Initial Costs:** Upfront investment for software, hardware, and infrastructure.
- **Maintenance and Upkeep:** Ongoing costs for system updates and technical support.
- **Training Costs:** Expense of training staff to use the system.
- **Operational Efficiency:** How the system reduces operational costs and improves workflow.
- **Return on Investment (ROI):** Long-term financial benefits like improved patient care and reduced errors.
- **Cost Savings:** Identifying areas where the system saves money (e.g., staffing, inventory).

2.4.3 Operational Feasibility:

- Operational feasibility assesses how well the proposed system will function within the existing operational processes and how effectively it solves identified problems. Below are the key points:
 - **Ease of Use:** The system will be user-friendly, allowing doctors, patients, and admin to navigate and perform tasks like booking appointments,

managing patient records, and prescribing medications without extensive training.

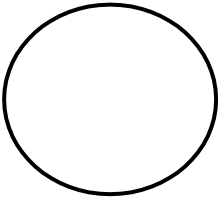
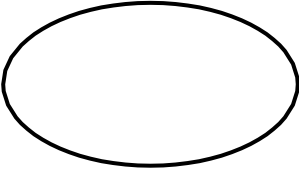
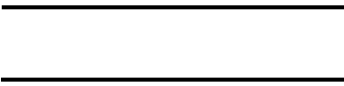
- **Improved Efficiency:** It will streamline hospital operations by automating processes like appointment scheduling, patient record management, and communication between departments, reducing manual errors and delays.
- **Compatibility with Current Operations:** The system fits well with the hospital's existing workflow. It will easily work with how doctors, and admin currently do their jobs, without causing any interruptions or major changes. The system will help make tasks like scheduling appointments, managing patient records, and communicating between departments smoother and more efficient, without disrupting day-to-day activities.
- **Scalability:** The system is designed to grow with the hospital's needs, making it capable of handling more patients, doctors, as the hospital expands, ensuring long-term operational sustainability.

3.1 Data Flow Diagram

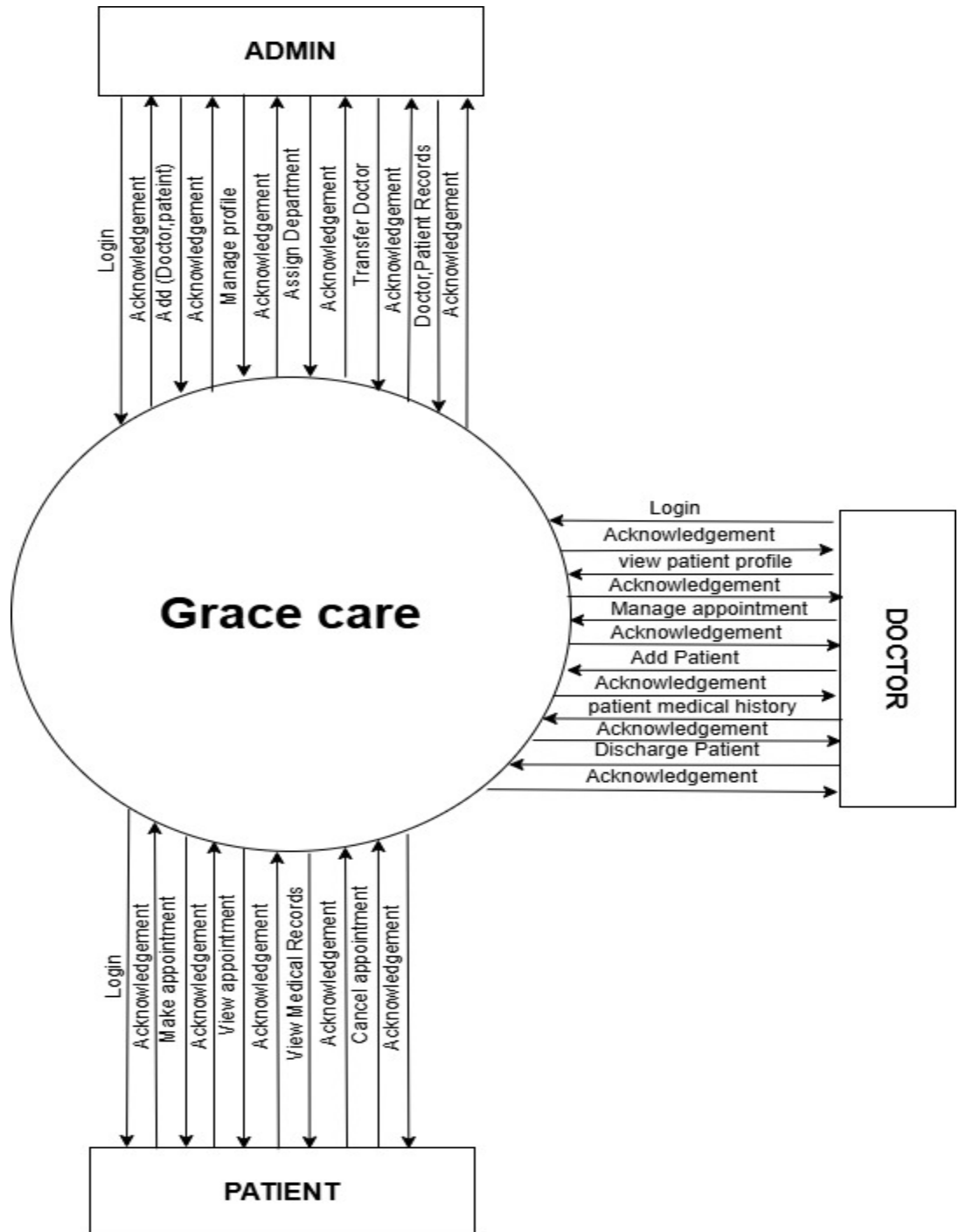
- DFD (data flow diagram) is also known as bubble chart or data flow graph.
- DFD's are very useful in understanding the system and can be effectively used during analysis. It shows flow of data through a system visually. The DFD is a hierarchical graphical model of a system the different processing activities or functions that the system performs and the data interchange among these functions.
- It views a system as a function that transforms the inputs into desired output.
- Each function is considered as a process that consumes some input data and produces some output data.
- Function model can be represented using DFD.
- DFD graphically representing the functions, or processes, which capture, manipulate, store, and distribute data between a system and its environment and between components of a system.
- The visual representation makes it a good communication tool between User and System designer.
- Structure of DFD allows starting from a broad overview and expand it to a hierarchy of detailed diagrams.

- DFD has often been used due to the following reasons:
 - Logical information flow of the system.
 - Determination of physical system construction requirements.
 - Simplicity of notation.
 - Establishment of manual and automated systems requirements.

[Table 2: Data Flow Diagram Symbols]

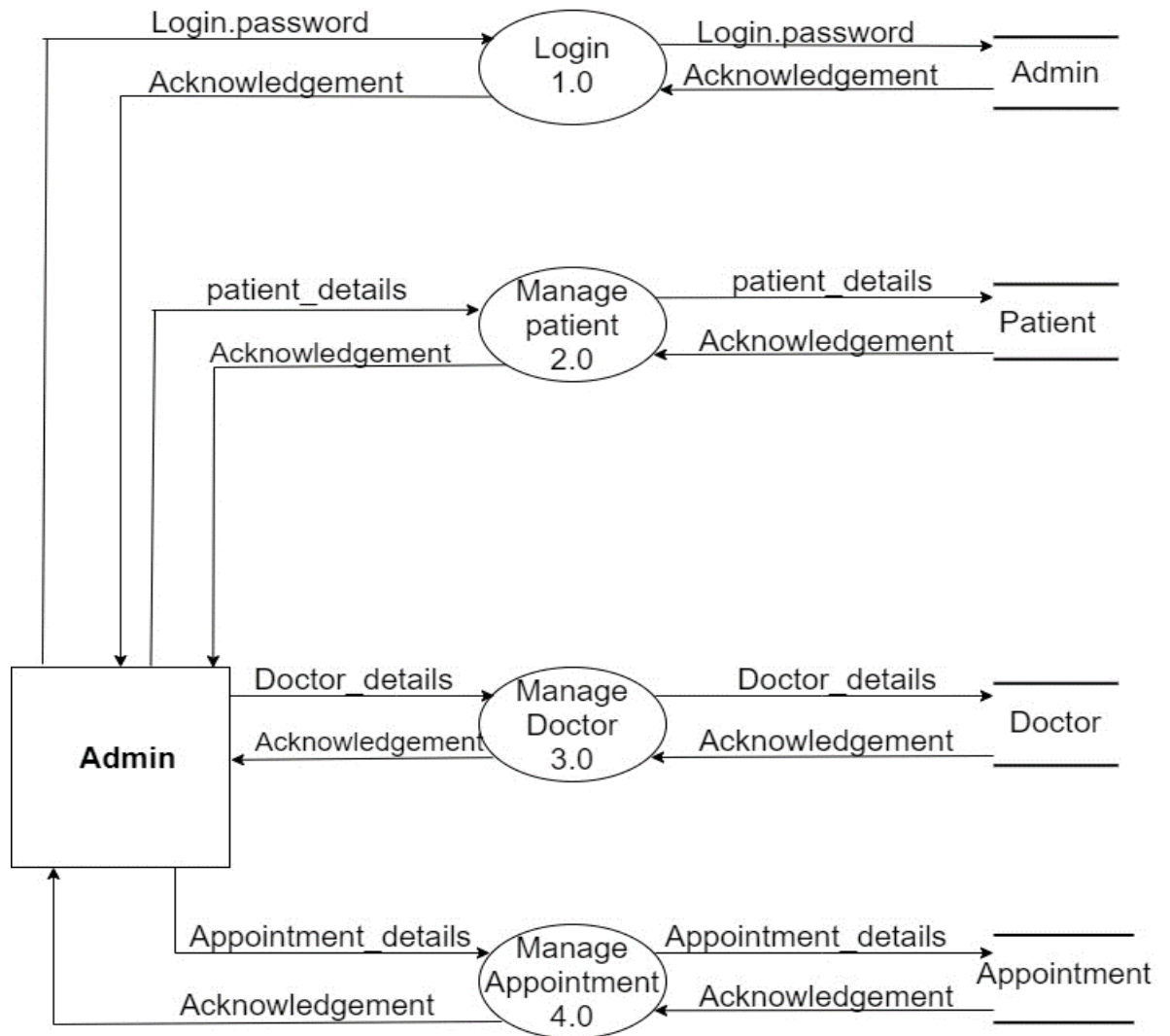
Symbols	Description
	Entity: Entities are external to the system which interacts by inputting the data.
	System: It shows the system name.
	Process: It shows the part of the system that transforms into outputs.
	Data Flow: It passes the data from one part to another.
	Data Store: Data store is represented by two parallel lines. It is generally logical file or database.

Level 0: Context



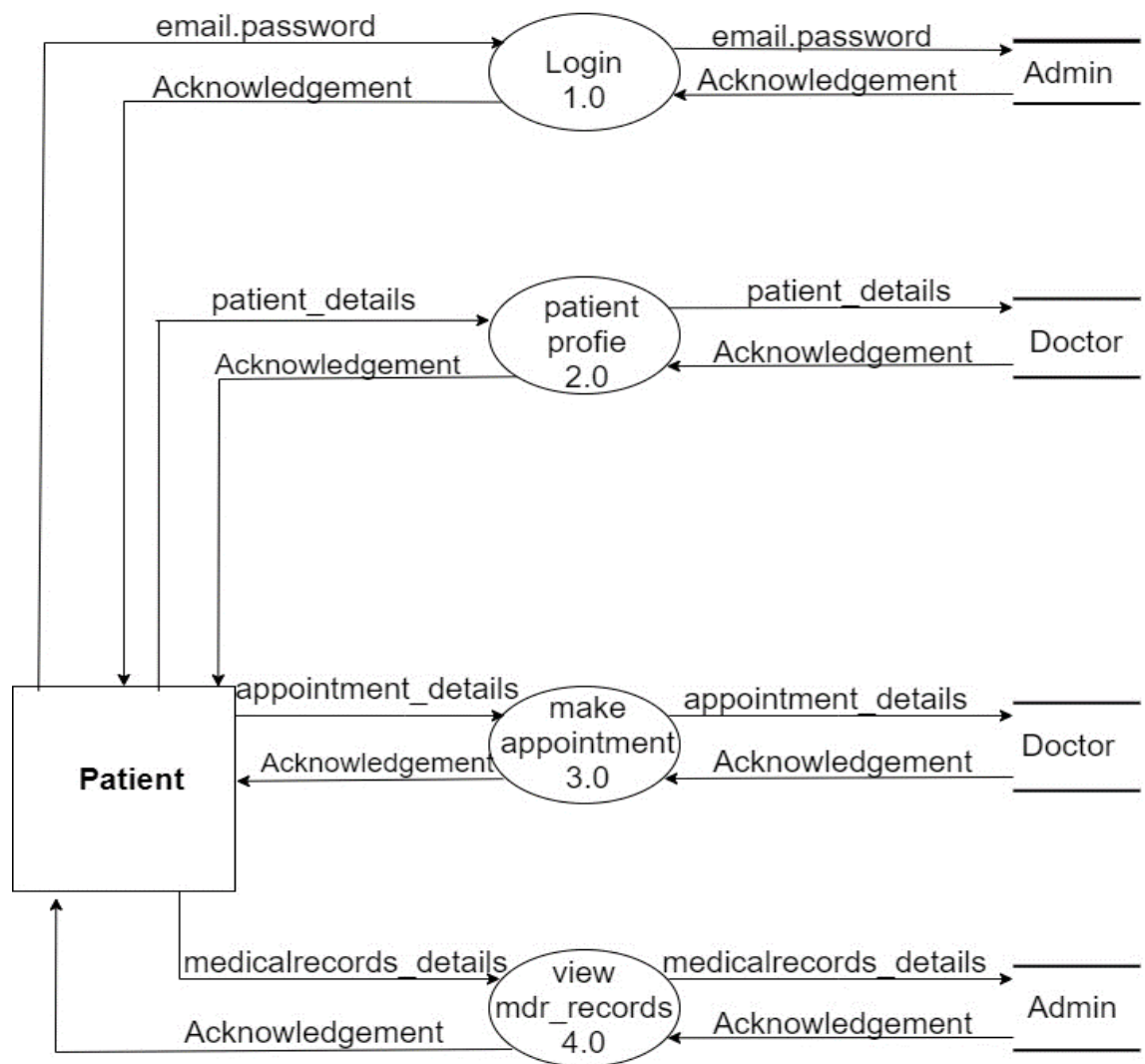
[Figure 2: Context level]

Level 1: Admin



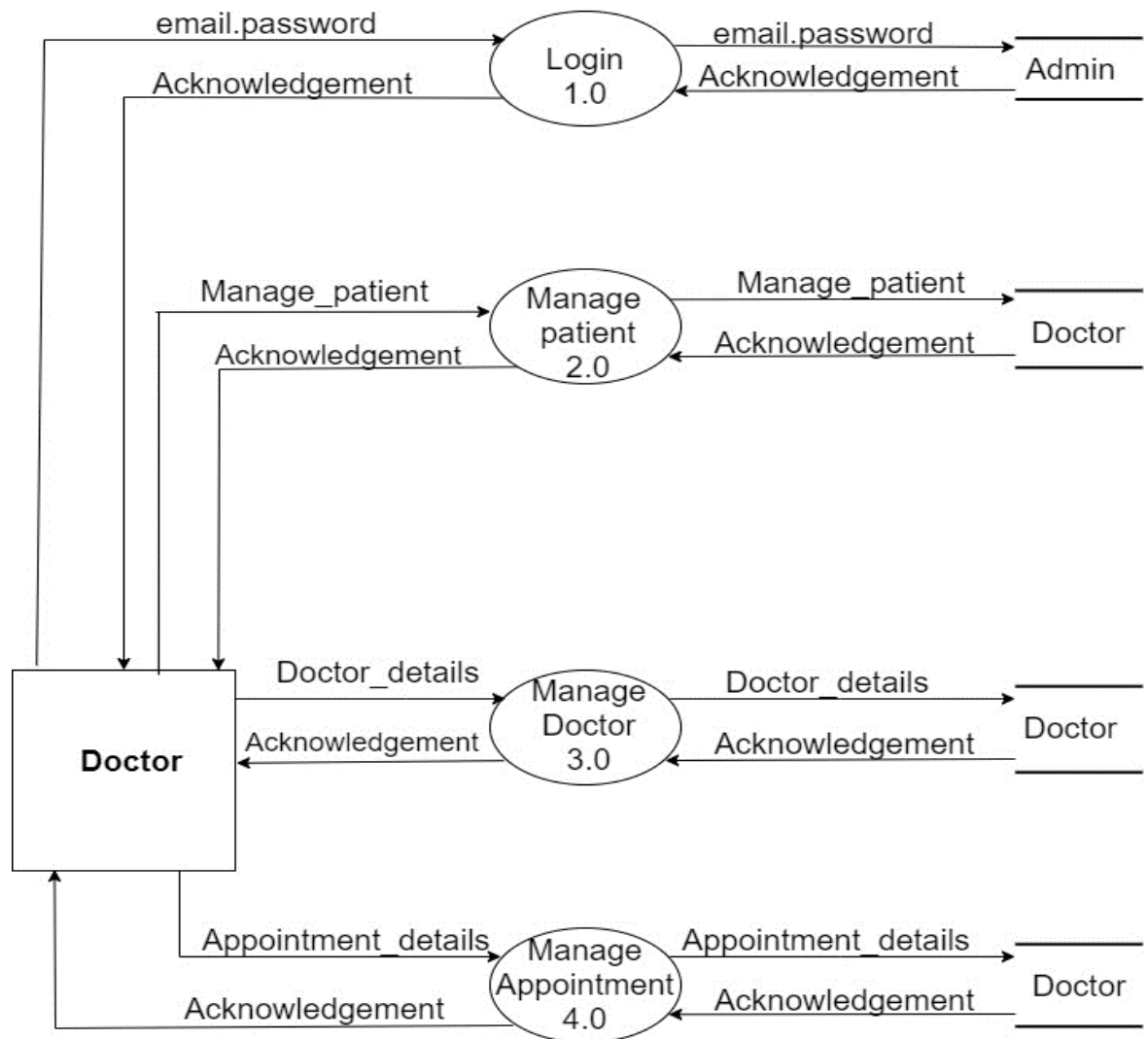
[Figure 3: DFD Level 1: Admin]

Level 1: Patient



[Figure 4: DFD Level 1: Patient]

Level 1: Doctor

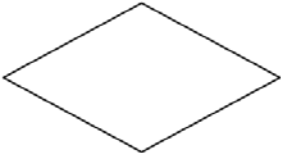
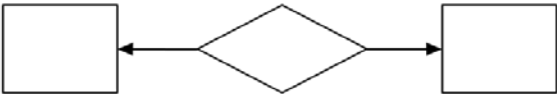
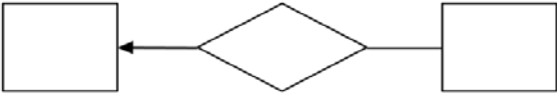
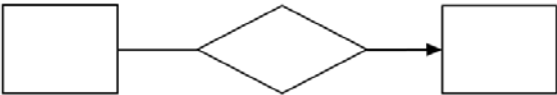
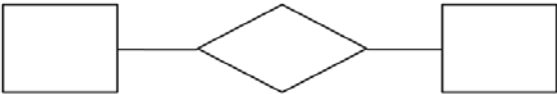


[Figure 5: DFD Level 1: Doctor]

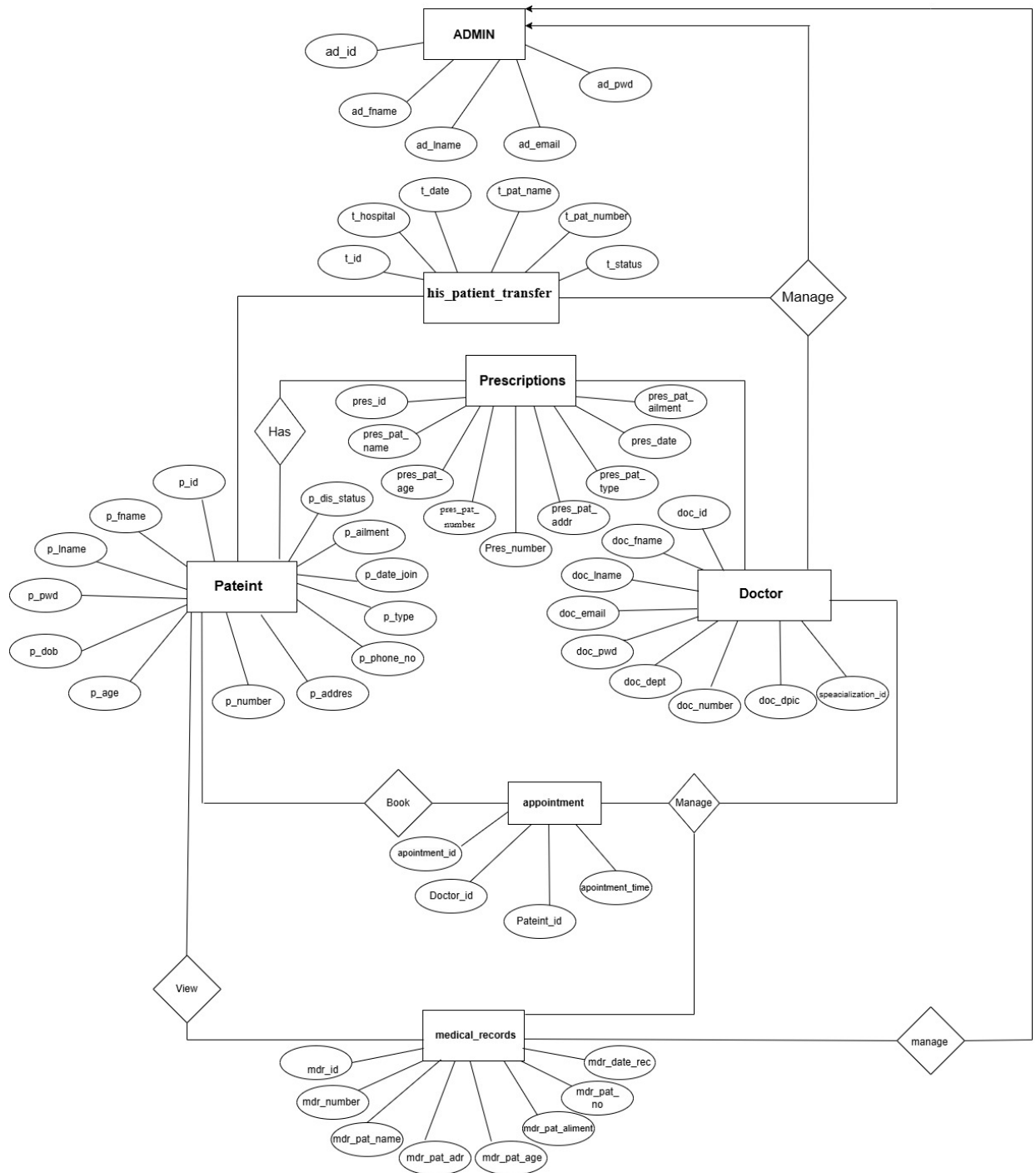
3.2 ER-Diagram

- An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research. Also known as ERDs or ER Models, they use a defined set of symbols such as rectangles, diamonds, ovals and connecting lines to depict the interconnectedness of entities, relationships and their attributes. They mirror grammatical structure, with entities as nouns and relationships as verbs.

[Table 3: ER-Diagram Symbols]

Symbols	Description
	Entity: Data object is real world entity or thing. It is represented by a rectangle shape. An entity is an object or concept about which you want to store information.
	Attributes: An attribute is property of characteristic of an entity. It is represented by oval shape.
	Relationship: Entity are connected each other via relations. Generally, relationships in binary because there are two entities are related to each other.
	Cardinality (One to One): An instance of entity A can relate to one instances of entity B.
	Cardinality (One to Many): An instance of entity A can relate to one or many instances of B but we can only relate one instance of A.
	Cardinality (Many to One): One or more instances of entity A can relate to one instances of B.
	Cardinality (Many to Many): One or more instances of entity A can relate to one more instance of entity B.

ER-Diagram:



[Figure 6: ER Diagram]

CHAPTER-4

SYSTEM MODELING

4.1 Database Dictionary

1. Table Name: Admin

Primary Key: id

[Table 4: Admin]

Field Name	Data Type (with size)	Constraints	Sample Data	Description
ad_id	Int (11)	Primary Key	1	Admin's id.
ad_fname	Varchar(100)	Not null	System	Admin's first name
ad_lname	Varchar(100)	Not null	Admin	Admin's last name
ad_email	Varchar(100)	Not null	admin@gmail.com	Admin's email username
ad_pwd	Varchar(100)	Not null	admin@123	Password of admin
ad_dpik	Varchar(100)	Not null	doc-icon.png	Admin's picture

2. Table Name: Doctor

Primary Key: id

Foreign Key: specialization_id

[Table 5: Doctor]

Field Name	Data Type (with size)	Constraints	Sample Data	Description
doc_id	Int(11)	Primary Key	1	Id of Doctor
doc_fname	Varchar(100)	Not null	new	Doctor's first name
doc_lname	Varchar(100)	Not null	doctor	Doctor's last name
doc_email	Varchar(100)	Not null	newdoctor@ gmail.com	Doctor's email
doc_pwd	Varchar(100)	Not null	NewDoctor1 23..	Doctor's passwoed
doc_dept	Varchar(100)	Not null	Doctor	Doctor's department
doc_numbe r	Varchar(100)	Not null	WAL3D	Doctor's number
doc_dpic	Varchar(100)	Not null	Photo.jpeg	Doctor's photo
specializati on_id	Int(11)	Foreign Key	2	Doctor's specializatio n id

3. Table Name : Patient

Primary Key: pat_id

[Table 6: Patient]

Field Name	Data Type (with size)	Constraints	Sample Data	Description
pat_id	Int(11)	Primary Key	1	Patient's id
pat_fname	Varchar(100)	Not null	new	Patient's first name
pat_lname	Varchar(100)	Not null	patient	Patient's last name
pat_password	Varchar(100)	Not null	newpatient@123.	Patient's password
pat_dob	Varchar(100)	Not null	05/01/01	Patient's date of birth
pat_age	Varchar(100)	Not null	22	Patient's age
pat_number	Varchar(100)	Not null	30USD	Patient's number
pat_addr	Varchar(100)	Not null	Ahmedabad	Patient's address
pat_phone	Varchar(100)	Not null	0987654321	Patient's mobile number
pat_type	Varchar(100)	Not null	Inpatient	Patient type

pat_date_ joined	Timestamp(6)	Not null	2024-04-06 20:00:08.2921 88	Patient's date of joined
pat_ailment	Varchar(100)	Not null	Fever	Patient's disease
pat_discharge status	Varchar(100)	Not null	Discharged	Patient's discharge status

4. Table Name : Appointment

Primary Key: appointment_id

Foreign Key: Doctor_id, Patient_id

[Table 7: Appointment]

Field Name	Data Type (with size)	Constraints	Sample Data	Description
appointment_id	Int(11)	Primary Key	1	Appointment id
doctor_id	Int(11)	Foreign Key	3	Doctor id
patient_id	Int(11)	Foreign Key	2	Patient id
appointment_time	datetime	Not null	2024-04-07 19:26:00	Appointment time
app_status	varchar(50)	Not null	approved	App status

5. Table Name : Medical records

Primary Key: mdr_id

[Table 8: Medical records]

Field Name	Data Type (with size)	Constraints	Sample Data	Description
mdr_id	Int(11)	Primary Key	1	Medical records id
mdr_number	varchar(200)	Not null	234	Medical records number
mdr_pat_name	varchar(200)	Not null	New	Medical records patient name
mdr_pat_adr	varchar(200)	Not null	Ahmedabad	Medical records patient address
mdr_pat_age	varchar(200)	Not null	25	Medical records patient age
mdr_pat_ailment	varchar(200)	Not null	fever	Medical records patient ailment
mdr_pat_number	varchar(200)	Not null	3ABCD	Medical records patient number
mdr_pat_prescr	long text	Not null	Dolo,metcin	Medical records patient prescription
mdr_date_rec	timestamp	Not null	2024-04-06 20:00:08.292188	Medical records date

6. Table Name : Prescriptions

Primary Key: pres_id

[Table 9: Prescriptions]

Field Name	Data Type (with size)	Constraints	Sample Data	Description
pres_id	Int(11)	Primary Key	1	Prescriptions id
pres_pat_ name	varchar(200)	Not null	New Patient	Prescriptions patient name
pres_pat_ age	varchar(200)	Not null	25	Prescriptions patient age
pres_pat_ number	varchar(200)	Not null	30USD	Prescriptions patient number
Pres_numbe r	varchar(200)	Not null	1FYN3	Prescriptions number
pres_pat_ addr	varchar(200)	Not null	AHMEDBA D	Prescriptions patient address
pres_pat_ type	varchar(200)	Not null	InPatient	Prescriptions patient typ
pres_date	timestamp(4)	Not null	2024-04-07 18:07:02.749 700	Prescriptions date
pres_pat_ ailment	varchar(200)	Not null	Fever	prescription aliment

7. Table Name : his_patient_transfer

Key: t_id

[Table 10: his_patient_transfer]

Field Name	Data Type (with size)	Constrain ts	Sample Data	Description
t_id	Int(20)	Primary Key , Auto_Increment	1	Transfer_id
t_hospital	Varchar(200)	Null	shelby	Hospital_name
t_date	Int(11)	Null	2024-04-07	Transfer_date
t_pat_name	varchar(200)	Null	harish	Patient_name
t_pat_number	varchar(200)	Null	ECJGI	Patient _Number
t_status	varchar(200)	Null	Success	Transfer Status

8. Table Name : Specializations

Primary Key: specialization_id

[Table 10: Specialization]

Field Name	Data Type (with size)	Constraints	Sample Data	Description
specialization_id	Int(11)	Primary Key	1	Specialization id
Specialization_ name	varchar(200)	Not null	Dentist	Specialization name

5.1 Hardware Specification:

5.1.1 Ram: 4GB.

5.1.2 Hard drive Storage needed: 10 GB.

5.1.3 Other Hardware Requirements: None.

5.2 Platform:

5.2.1 Supported Operating System: Windows and above LINUX and Macos is compatible.

5.2.2 Programmer Server: Apache Server 2.2.

5.3 Framework:

5.3.1 Mark-up Language: HTML4 and HTML5.

5.3.2 Programming Language: PHP 5.3.

5.4 Technical Specification:

5.4.1 Front-End: PHP 5.3.

5.4.2 Back-End: MySQL 5.5.24.

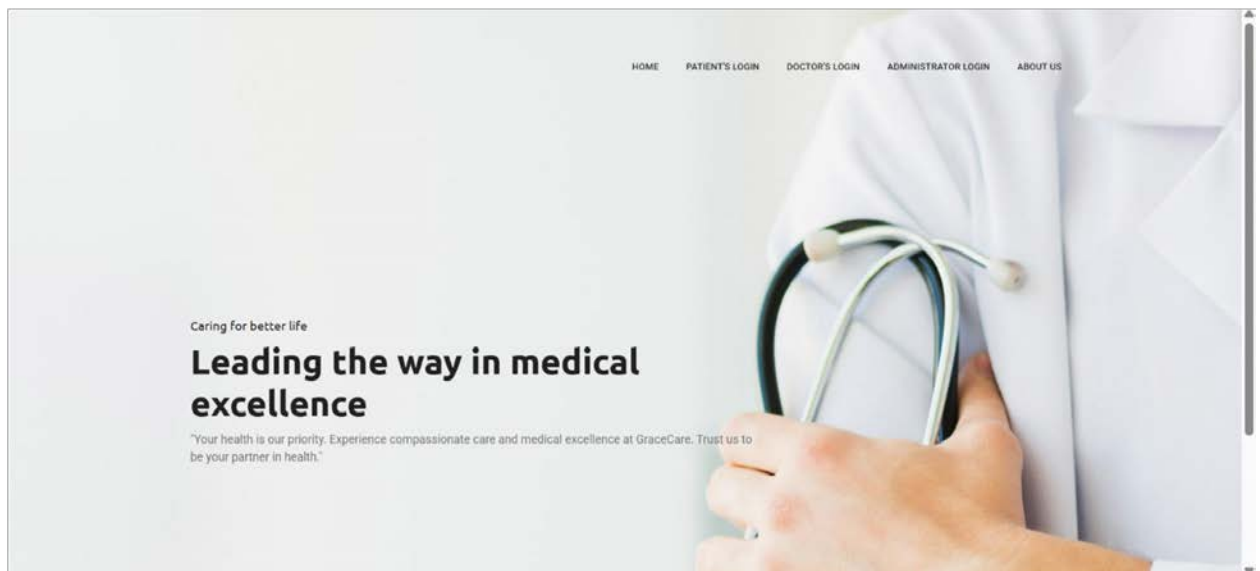
5.4.3 IDE Tools: Sublime Text3 and Marco Media Dreamweaver.

5.4.4 UML Tools: Microsoft Office Visio 2007.

5.4.5 SRS Tools: Microsoft Word 2016.

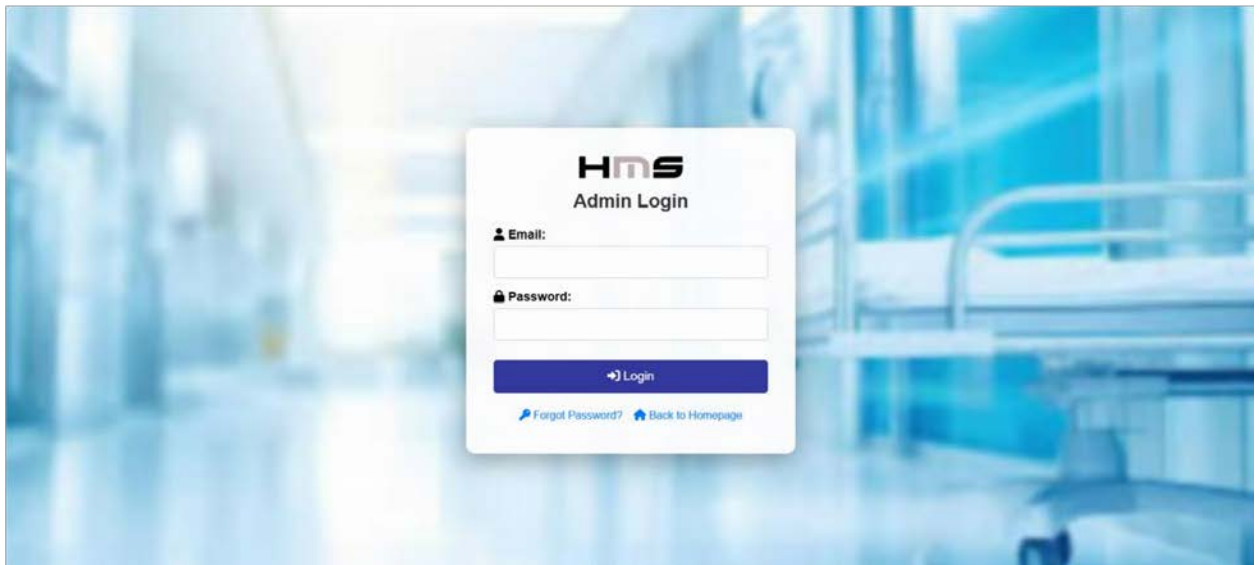
5.5 DESIGN LAYOUT

Homepage:-

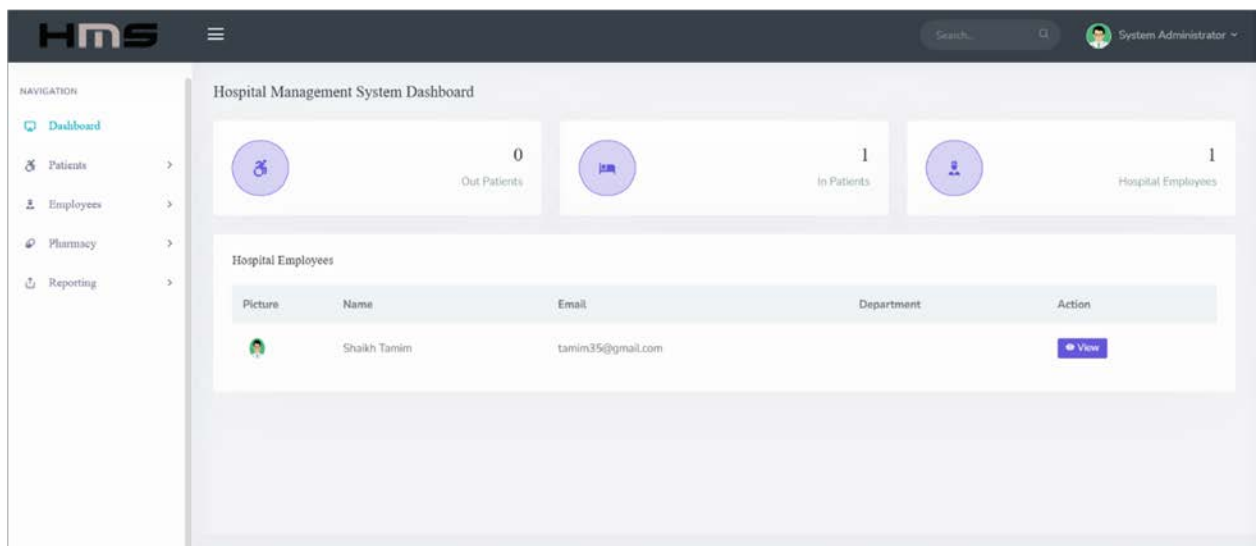


[Figure: Homepage]

Admin:-

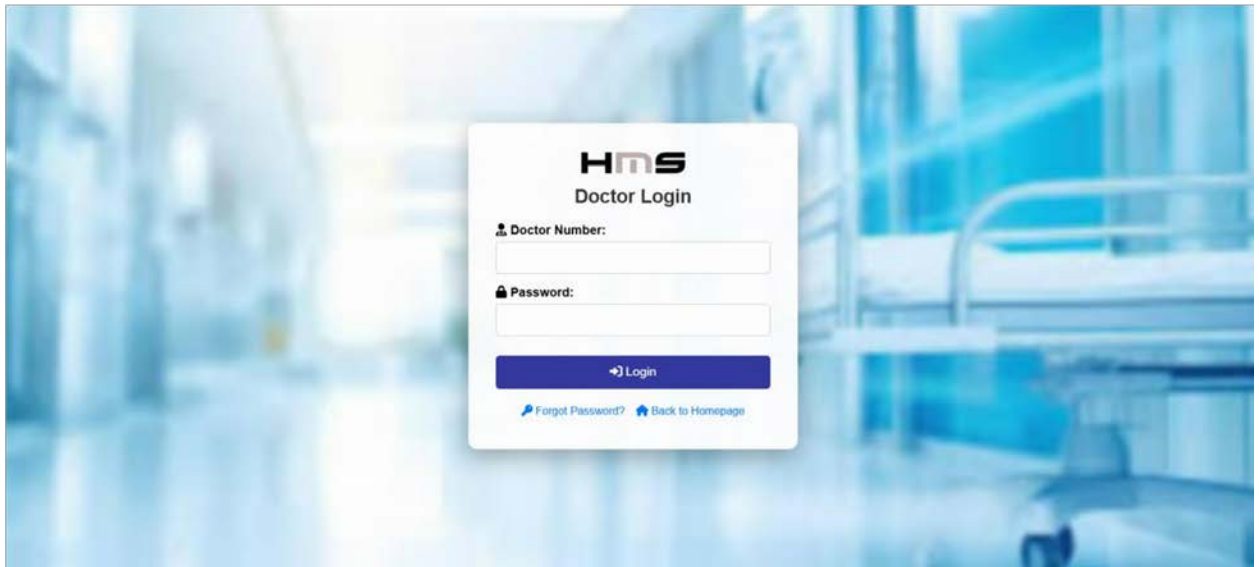


[Figure: Admin page]

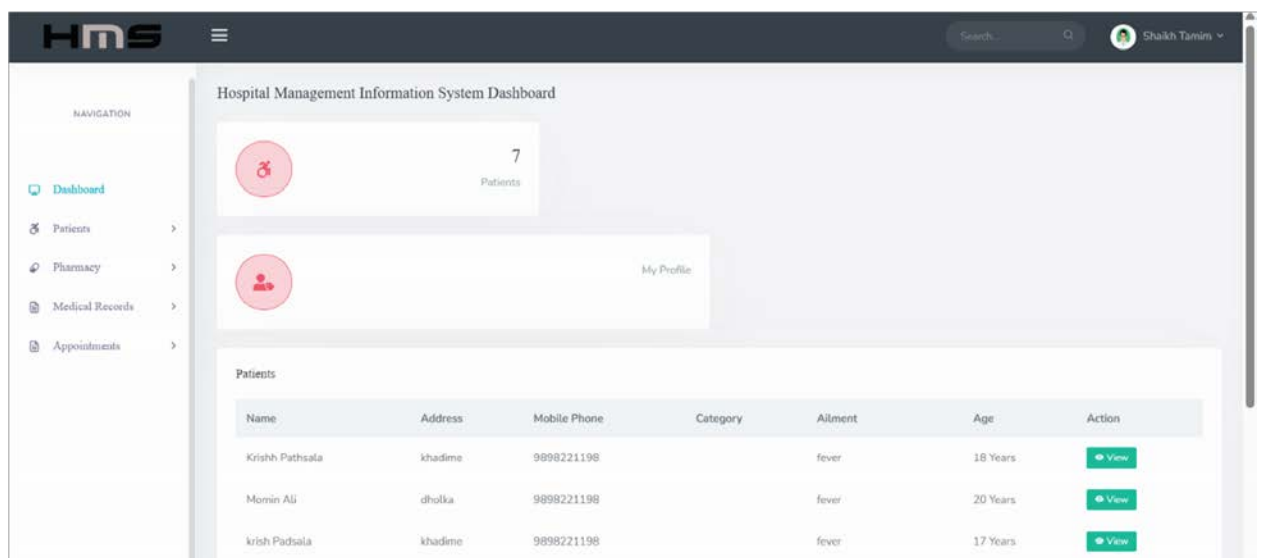


[Figure: Admin Dashboard]

Doctor:-



[Figure: Doctor Login]



[Figure: Doctor Dashboard]

HMS Search... Shaikh Tamim

NAVIGATION

- Dashboard
- Patients**
 - Register Patient
 - View Patients
 - Manage Patients
- Discharge Patients
- Patient Transfers
- Pharmacy
- Medical Records
- Appointments

Add Patient Details Dashboard > Patients > Add Patient

Fill all fields

First Name: Patient's First Name Last Name: Patient's Last Name

Set Password:

Date Of Birth: dd-mm-yyyy Age: Patient's Age

Address: Patient's Address

Mobile Number: Patient Ailment: Patient's Type: Choose

Add Patient

[Figure: Add Patient Details]

HMS Search... Shaikh Tamim

NAVIGATION

- Dashboard
- Patients**
 - Register Patient
 - View Patients
 - Manage Patients
- Discharge Patients
- Patient Transfers
- Pharmacy
- Medical Records
- Appointments

Patient Details Dashboard > Patients > View Patients

Search

#	Patient Name	Patient Number	Patient Address	Patient Phone	Patient Age	Patient Category	Action
1	Anish Saiyed	ABCD9	khadime	9898221198	18 Years		View
2	Krishh Pathsala	ECJGT	khadime	9898221198	18 Years		View
3	Harish Shaikh	ECJGI	khadime	9898221121	24 Years	InPatient	View
4	krish Padsala	ABCD6	khadime	9898221198	17 Years		View
5	krish padsala	AVCD	khadime	9898221198	18 Years		View
6	Shaikh Yamin	ASDI1	khadime	9898221198	16 Years		View
7	Momin Ali	E4YT3	dholka	9898221198	20 Years	InPatient	View

1

[Figure: View Patient Details]

HMS

Search...

Shaikh Tamim

NAVIGATION

Dashboard

Patients

Register Patient

View Patients

Manage Patients

Discharge Patients

Patient Transfers

Pharmacy

Medical Records

Appointments

Manage Patient Details

Dashboard > Patients > Manage Patients

Search

#	Patient Name	Patient Number	Patient Address	Patient Category	Action
1	Anish Salyed	ABCD9	khadime		View Update
2	Momin Ali	E4YT3	dholka	InPatient	View Update
3	krish Padsala	ABCD6	khadime		View Update
4	Shaikh Yamin	ASDJ1	khadime		View Update
5	krish padsala	AVCD	khadime		View Update
6	Krishh Pathsala	ECJGT	khadime		View Update
7	Harish Shaikh	ECJGI	khadime	InPatient	View Update

1

[Figure: Manage Patient Details]

HMS

Search...

Shaikh Tamim

NAVIGATION

Dashboard

Patients

Register Patient

View Patients

Manage Patients

Discharge Patients

Patient Transfers

Pharmacy

Medical Records

Appointments

Discharge Patients

Dashboard > Patients > Discharge Patients

Search

#	Patient Name	Patient Number	Patient Address	Patient Category	Action
1	Harish Shaikh	ECJGI	khadime	InPatient	Discharge
2	Anish Saiyed	ABCD9	khadime		Discharge
3	Shaikh Yamin	ASDJ1	khadime		Discharge
4	Krishh Pathsala	ECJGT	khadime		Discharge
5	Momin Ali	E4YT3	dholka	InPatient	Discharge
6	krish padsala	AVCD	khadime		Discharge
7	krish Padsala	ABCD6	khadime		Discharge

1

[Figure: Discharge Patient]

HMS

Search...

Shaikh Tamim

NAVIGATION

Dashboard

Patients

Register Patient

View Patients

Manage Patients

Discharge Patients

Patient Transfers

Pharmacy

Medical Records

Appointments

Transfer Patients

Dashboard > Patients > Transfer Patients

Patient's Awaiting Transfers

Search

#	Patient Name	Patient Number	Patient Address	Patient Category	Action
1	Harish Shaikh	ECJGI	khadime	InPatient	Transfer Patient
2	Momin Ali	E4YT3	dholka	InPatient	Transfer Patient

1

Transferred Patients

Search

#	Patient Name	Transfer Number	Transfer Status	Referral Hospital	Transfer Date
1	NEW PATIENT	30USD	Success	NewHospital	2024-04-06

[Figure: Transfer Patient]

HMS

Search...

Shaikh Tamim

NAVIGATION

Dashboard

Patients

Pharmacy

Add Prescriptions

View Prescriptions

Manage Prescriptions

Medical Records

Appointments

Add Prescriptions

Dashboard > Pharmacy > Give Prescription

Search

#	Patient Name	Patient Number	Patient Address	Patient Ailment	Patient Age	Patient Category	Action
1	krish Padsala	ABCD6	khadime	fever	17 Years		Add Prescription
2	Harish Shaikh	ECJGI	khadime	Skin Reaction	24 Years	InPatient	Add Prescription
3	krish padsala	AVCD	khadime	fever	18 Years		Add Prescription
4	Krishh Pathsala	ECJGT	khadime	fever	18 Years		Add Prescription
5	Momin Ali	E4YT3	dholka	fever	20 Years	InPatient	Add Prescription
6	Anish Saiyed	ABCD9	khadime	fever	18 Years		Add Prescription
7	Shaikh Yamin	ASDI1	khadime	fever	16 Years		Add Prescription

1

[Figure: Add Prescription]

HMS

Search...

Shaikh Tamim

NAVIGATION

Dashboard

Patients

Pharmacy

Add Prescriptions

View Prescriptions

Manage Prescriptions

Medical Records

Appointments

Manage Prescriptions

Dashboard Pharmacy Manage Prescriptions

Search

#	Patient Name	Patient Number	Address	Ailment	Age	Category	Action
1	yamin shaikh	H7S53	AHMEDBAD	HELLO	22 Years	InPatient	View Update
2	NEW PATIENT	30USD	AHMEDBAD	hello	25 Years	InPatient	View Update
3	yamin shaikh	H7S53	AHMEDBAD	HELLO	22 Years	InPatient	View Update
4	yamin shaikh	H7S53	AHMEDBAD	HELLO	22 Years	InPatient	View Update
5	Harish Shaikh	ECIGI	khadime	Skin Reaction	24 Years	InPatient	View Update
6	Harish Shaikh	ECIGI	khadime	Skin Reaction	18 Years	InPatient	View Update
7	Harish Shaikh	ECIGI	khadime	Skin Reaction	18 Years	InPatient	View Update

1 2

[Figure: Manage Prescription]

HMS

Search...

Shaikh Tamim

NAVIGATION

Dashboard

Patients

Pharmacy

Medical Records

Add Medical Record

Manage Medical Records

Appointments

Patient Details

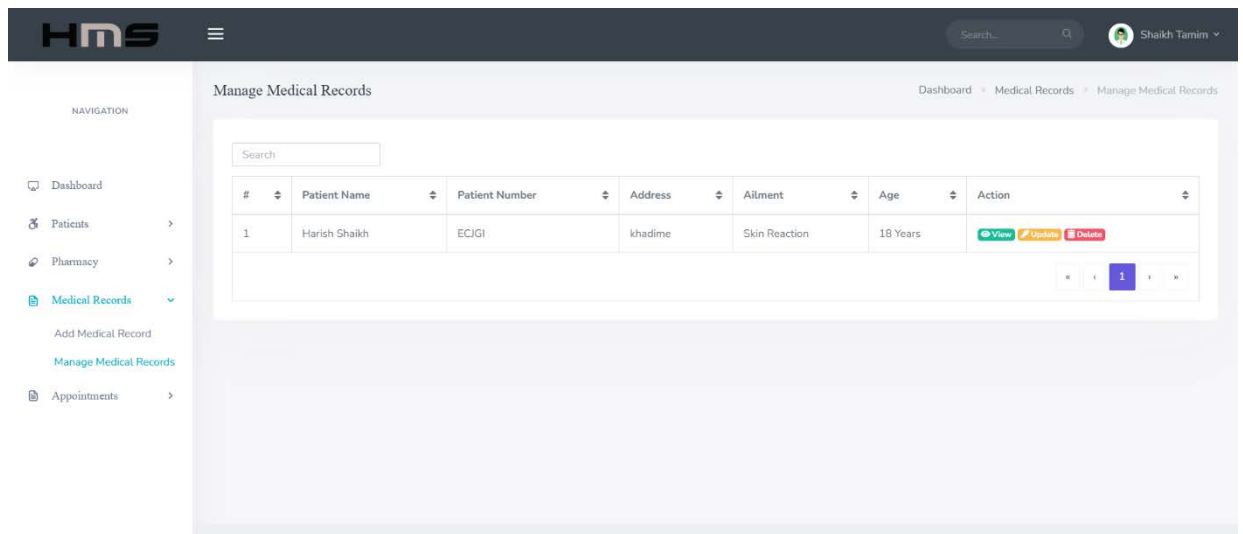
Dashboard Medical Records Add Medical Records

Search

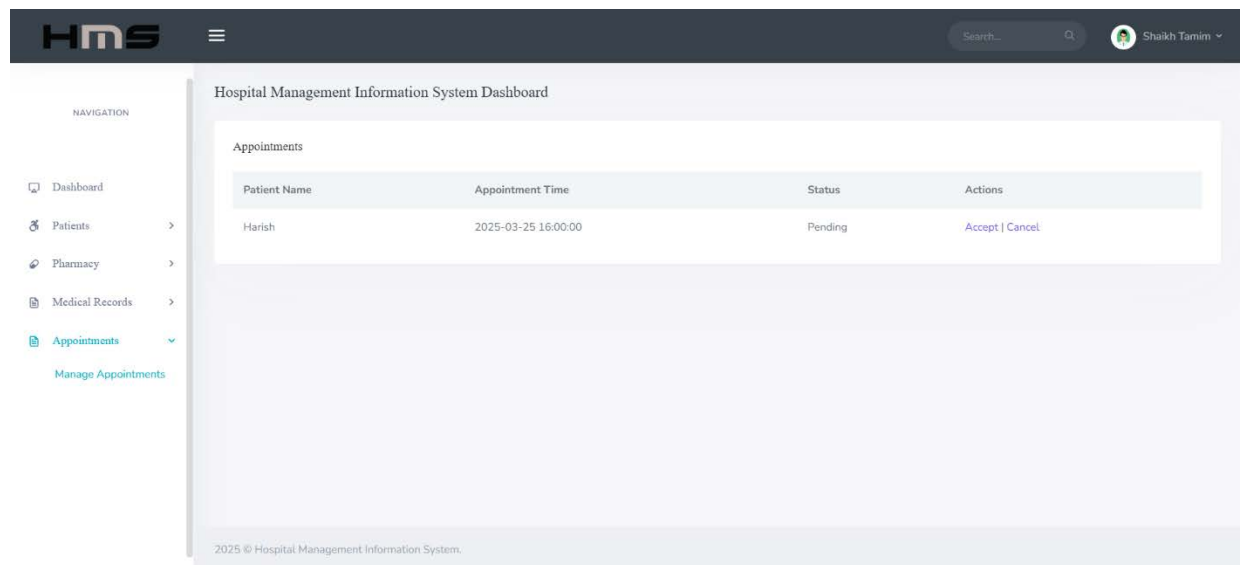
#	Name	Number	Address	Phone	Age	Category	Action
1	Shaikh Yamin	ASDJ1	khadime	9898221198	16 Years		Add Medical Record
2	Krishh Pathsala	ECIGT	khadime	9898221198	18 Years		Add Medical Record
3	krish Padsala	ABCD6	khadime	9898221198	17 Years		Add Medical Record
4	Harish Shaikh	ECIGI	khadime	9898221121	24 Years	InPatient	Add Medical Record
5	krish padsala	AVCD	khadime	9898221198	18 Years		Add Medical Record
6	Anish Saiyed	ABCD9	khadime	9898221198	18 Years		Add Medical Record
7	Momin Ali	E4YT3	dholka	9898221198	20 Years	InPatient	Add Medical Record

1

[Figure: Add Medical Records]

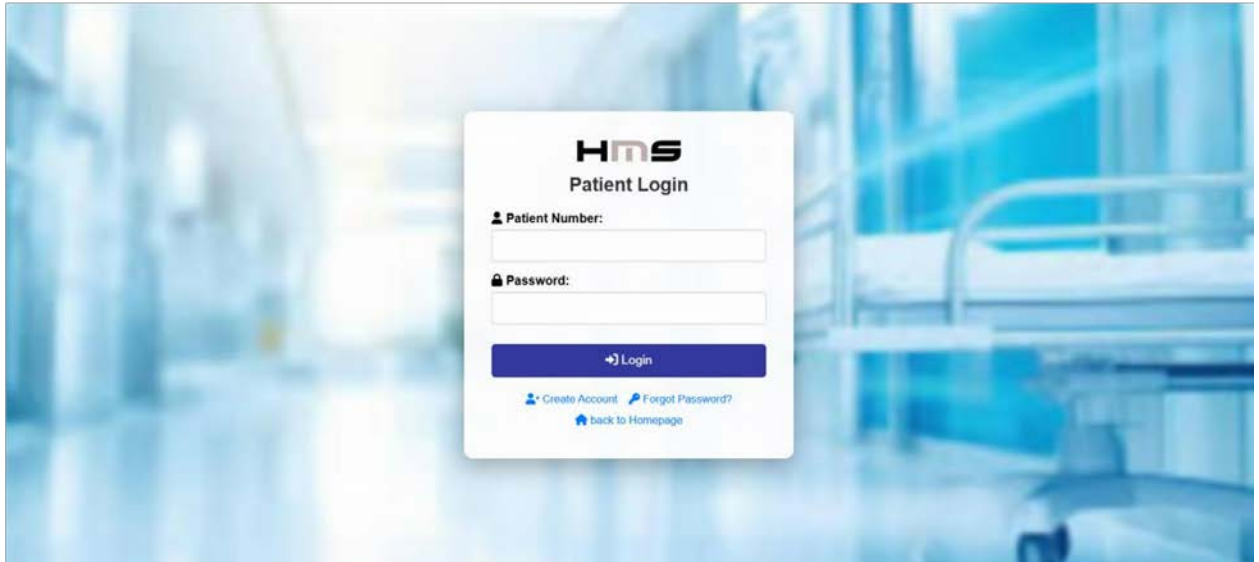


[Figure: Manage Medical Records]

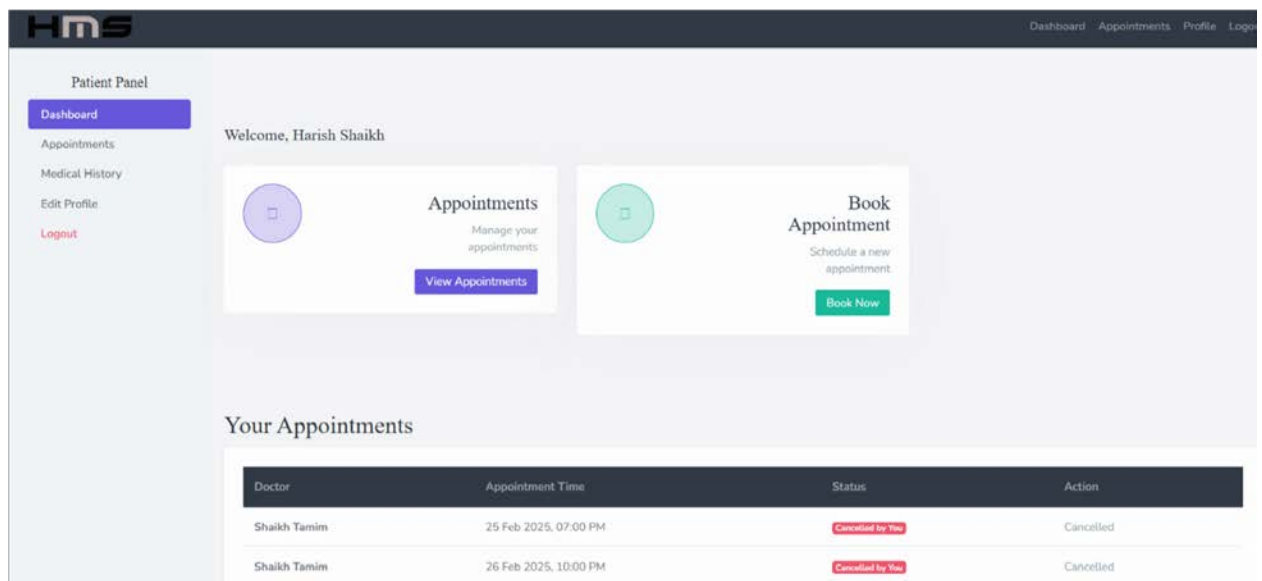


[Figure: Manage Appointment]

Patient:-



[Figure: Patient login]



[Figure: Patient Dashboard]

Appointment:-



Book an Appointment

Doctor:

Date:

Time:

[Confirm Booking](#) [Cancel](#)

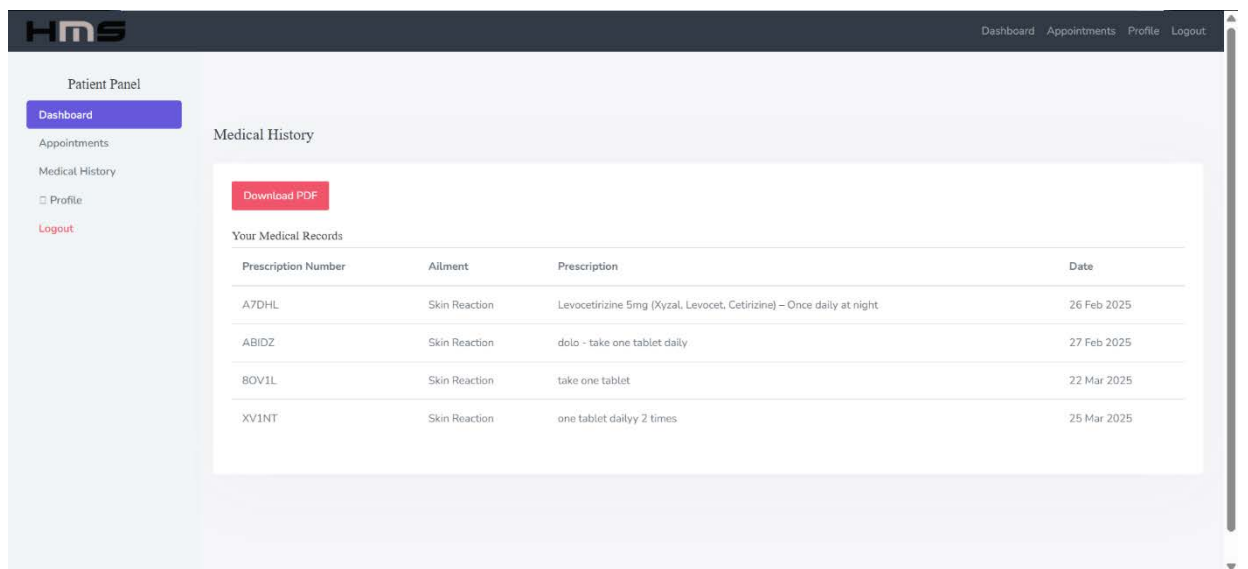
[Figure: Book Appointment]

Your Appointments

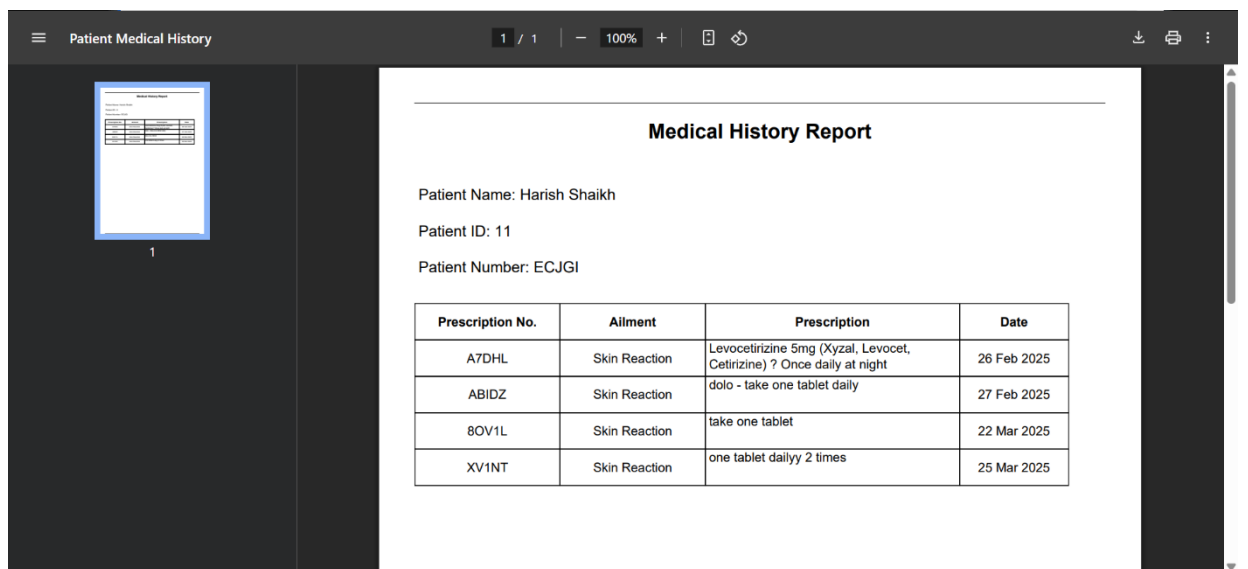
Doctor	Appointment Time	Status	Action
Shaikh Tamim	2025-03-25 16:00:00	pending	Cancel

[Back to Dashboard](#)

[Figure: View Appointment]



[Figure: Medical History]



[Figure: Medical History Report]

CHAPTER 6

TESTING

6.1 Testing Methods

There are different models of testing. On the basis of testing methods there are two types of testing:

1. White-box testing
2. Black-box testing

Black-box tests are used to demonstrate that software functions are operationally, that input is properly accepted and output is correctly produced, and that integrity of external information is maintained.

White-box tests are used to examine the procedural details. It checks the logical paths by test case. It can also check the conditions, loops used in the software coding. It checks that loops are working correctly on defined boundary value.

6.1.1. WHITE-BOX TESTING

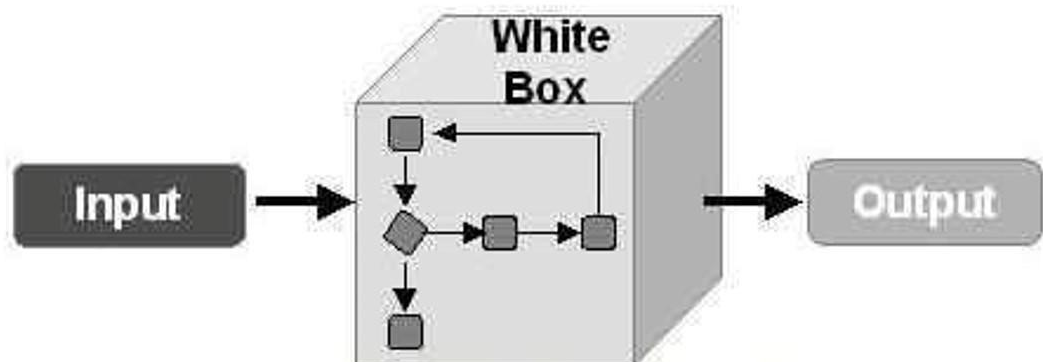
White-box testing sometimes called glass-box testing, is a test case design method that uses the control structure of the procedural design to drive the test case. Always we are thinking that there is no necessary to execute or check the loops and conditions. And so large number of errors is uncovered. With using white-box testing methods, we have

checked that, all possible ways within a function have been executed at least once.

All conditional statements have been evaluated for both true and false outcomes.

Each loop iteration operates correctly at its upper and lower boundary values, adhering to its defined conditions.

In our coding practices, we ensure thorough testing of all loops within each module. One specific method of white-box testing employed is basis path testing, which comprises two components: flow graph notation and cyclomatic complexity analysis. Flow graph notation facilitates examination of the logical control flow, while cyclomatic complexity aids in assessing the intricacy of our project's structure.



[Figure 8: White Box]

6.1.2. BLACK-BOX TESTING

The technique of testing without having any knowledge of the interior workings of the application is Black Box testing. The tester is oblivious to the system architecture and does not have access to the source code. Typically, when performing a black box test, a tester will interact with the system's user interface by providing inputs and examining outputs without knowing how and where the inputs are worked upon.

We used following problems in our coding to find errors in the following categories:

Incorrect or missing function.

Interface errors.

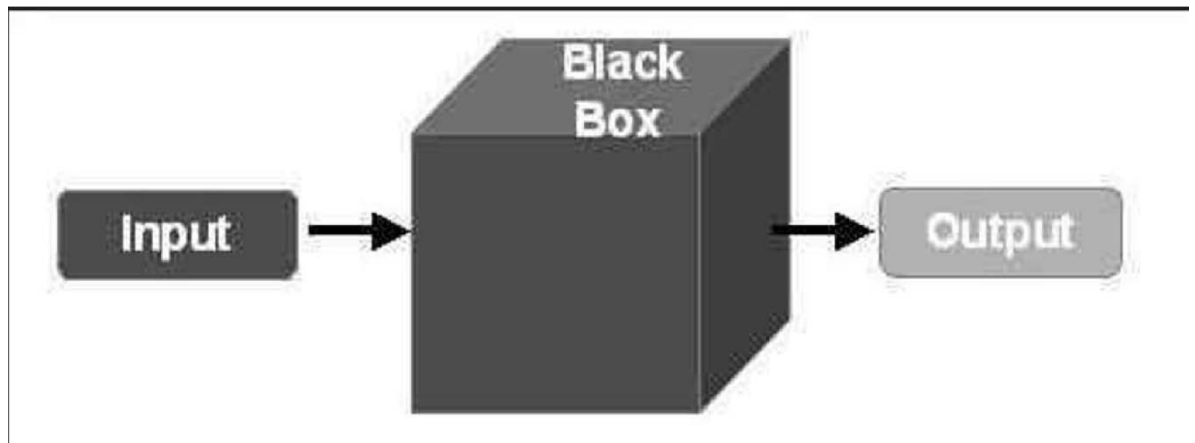
Error in database.

Performance errors.

Initialization and termination errors.

Unlike white-box testing, which is performed earlier in the testing process, black-box testing tends to be applied during later stages of testing.

By employing black-box methodologies, we formulate a series of test cases that meet the following criteria: Test cases are designed to diminish the need for additional testing by a count greater than one, ensuring efficient and thorough testing.



[Figure 9: Black Box]

CONCLUSION

In summary, the Hospital Management System, developed using agile methods, has greatly improved healthcare administration and patient care. By working together and adapting to changes, we've created a streamlined platform for managing patient information and administrative tasks. Moving forward, we'll keep improving the system to meet the needs of healthcare providers and patients, making hospital operations more efficient and effective.

BIBLIOGRAPHY

▪ BOOK REFERENCES

[1] PHP and MySQL Web Development, Fifth Edition Authors: Luke Welling and Laura Thomson Published by : Addison-Wesley Professional in 2016.

[2] The complete Reference PHP, Eight Edition Author: Steven Holzner Published by: Tata McGraw-Hill Edition 2008.

[3] Database Management System, Fifth Edition Author: Dr.Rajiv Chopra Published by: Chand & Company.

▪ WEB REFERENCES

[1] W3Schools: "HTML Tutorial"

Link: <https://www.w3schools.com/html/>

[2]CSS-Tricks: "A Complete Guide to Flexbox"

Link: <https://css-tricks.com/snippets/css/a-guide-to-flexbox/>

[3] Mozilla Developer Network (MDN):"JavaScript Basics"

Link: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>

[4] Geeks for geeks

Link: <https://www.geeksforgeeks.org>